

## LR(0) Parsing

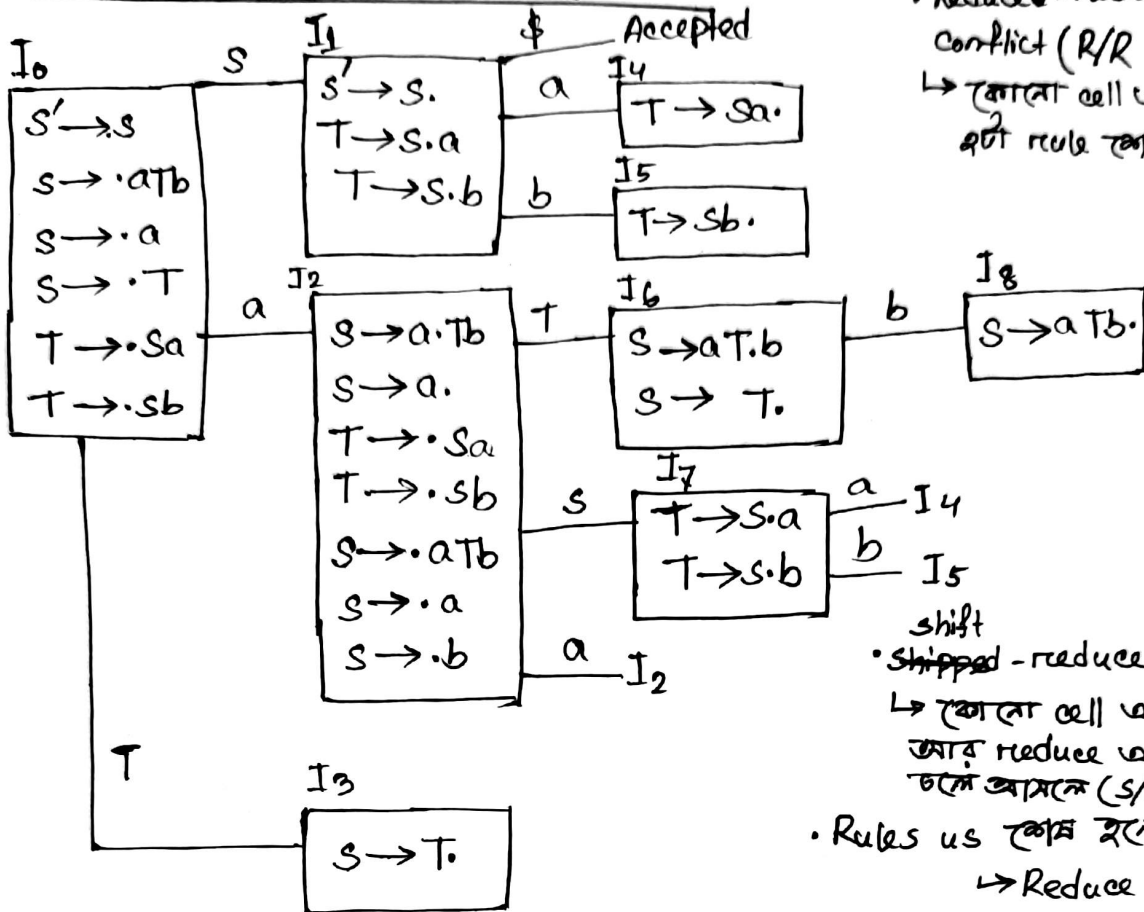
$$S \rightarrow aTb \mid a \mid T$$

$$T \rightarrow Sa \mid Sb$$

### Step-01 [Augmented Grammar]

0.  $S' \rightarrow S$
1.  $S \rightarrow aTb$
2.  $S \rightarrow a$
3.  $S \rightarrow T$
4.  $T \rightarrow Sa$
5.  $T \rightarrow Sb$

## Step-02 [Transition Diagram]



• Reduced - Reduced

Conflict (R/R)

→ কোন cell এ একসাথে দুটি rule কাজ দেন

• Rules us কাজ দেন

→ Reduce operation

$S \rightarrow$  shipped shift

$r \rightarrow$  rule

## Step-03 [LR(0) Parsing Table]

| State | Action Table |       |        | Go To |   |
|-------|--------------|-------|--------|-------|---|
|       | a            | b     | \$     | S     | T |
| 0     | S2           |       |        | 1     | 3 |
| 1     | S4           | S5    | Accept |       |   |
| 2     | S2/r2        | r2    | r2     | 7     | 6 |
| 3     | r3           | r3    | r3     |       |   |
| 4     | r4           | r4    | r4     |       |   |
| 5     | r5           | r5    | r5     |       |   |
| 6     | r3           | S8/r3 | r3     |       |   |
| 7     | S4           | S5    |        |       |   |
| 8     | r1           | r1    | r1     |       |   |

• If there is conflict (S/R or R/R) → then it won't follow LR(0)